

Opening Collateral Data and Small Firm Credit Access

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Research Proposal – Internal Use Only

Motivation

- Small business finance faces important collateral-related frictions
- Lenders **cannot verify** the assets small firms own, and even when they can, they **cannot enforce** claims on them
 - ⇒ Small firms hold productive assets but cannot borrow against them
- **Opening collateral data** addresses both frictions by making claims over assets publicly searchable and enforceable
- **This Project:** What are the effects of open collateral registries on small firm credit access and bank competition

Research question

How do open collateral registries affect small firm credit access and lender competition?

- What is the role of enforcing collateral recovery?
 - Registration converts the firm's assets into **usable collateral**
- What is the role of competition for borrowers who have verifiable collateral?
 - It converts a private lending relationship into a **market-wide signal** of borrower quality
- Whose information do lenders choose to share and how this shapes the effectiveness of the policy?
 - Trade-off **formal seniority** for the potential **erosion of informational rents**

Setting and Data

SIGM: A national movable collateral registry

- Peru created the **Sistema Informativo de Garantías Mobiliarias (SIGM)** in March 2025
 - National electronic registry of security interests over movable assets
- Registration is the **creditor's choice**, not a mandate
- Filing confers **priority and enforceability** against third parties ...
- ... but publishes the **borrower's identity**, the **collateral description**, and the **creditor's name** to a publicly searchable database
- Registration therefore trades formal seniority for the potential erosion of informational rents

Sources of heterogeneity

Lenders. Peru's credit market is segmented by **information technology**

- **Commercial banks** lend on hard information and verifiable collateral
- **Microfinance institutions** lend on soft, relationship-based information; public disclosure erodes rents

Courts. Seniority is only valuable where it can be executed

- Judicial pilot in the Corte Superior de Lima from April 2025
- October 2025: 18 Superior Courts connected; 16 courts still to follow
- Staggered variation in **local enforcement**

Firms. Reporting fixed collateral versus those with only movable or none

Data

- Peru's **Credit Registry (CR)**

Reporte Crediticio Consolidado

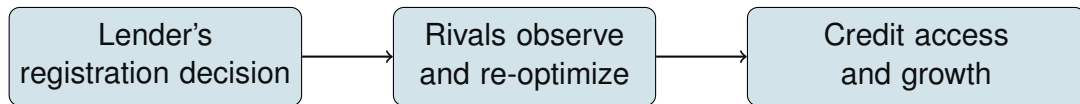
- Universe of outstanding debt for every borrower-lender pair
- Guarantee information, lender-reported due to regulation and provisions: movable vs. immovable, value
- Interest rate reports with loan-level information at origination for small business loans, micro-credit, and consumer loans
- Bank branch publicly available data for local competition measures
- Judicial rollout across Superior Courts

These datasets, if combined with registration records, will allow us to trace the effects of SIGM on credit access, interest rates, and small firm performance

Conceptual Framework and Predictions

Roadmap: three building blocks

1. **Disclosure incentives**: which lenders register which borrowers' collateral
2. **Competitive response**: how rivals react when they observe a registered collateral
3. **Borrower outcomes**: credit access, interest rates, and real activity



I. The registration decision

- Registering confers **enforceable seniority** against third-party claims
- Gains from registration increase with:
 - The **liquidation value** of the collateral
 - The likelihood that the borrower has **other creditors**
 - Quality of **enforcement**
- Cost of disclosure:
 - Largest for **relationship lenders (MFIs)**, whose rents derive from private knowledge of borrower quality
 - Scales up with **local competition**: registration is only dangerous when rivals observe and act on the information

Testable predictions

1. **Lender-type heterogeneity**: conditional on observables, banks register movable guarantees at higher rates than MFIs
2. **Within-portfolio**: registration concentrates on larger, multi-lender, formal firms whose informational rents are low, and is less likely among captive borrowers with high informational rents
3. **Local markets**: registration increases with local court efficiency and declines with local competitor presence

II. The competitive response and testable predictions

- A registered notice reveals a **screened borrower with enforceable collateral** and a rival can make a competing offer
 - Poaching intensity increases with the number of rivals in the borrower's city
 - The incumbent's hold-up rents erode as competition intensifies
- 1. **Poaching** concentrates on borrowers with registered movable guarantees and increases with rival branch density in the borrower's city
- 2. **Interest rates** fall for registered borrowers; most where rival presence is dense
- 3. **Within relationship-lender portfolios**: rates rise or credit contracts for the smallest, unsecured, opaque borrowers

III. Net effect on small business outcomes

- Two competing forces:
 - Gains from verifiable and enforceable collateral: contested borrowers receive cheaper, larger credit from new lenders
 - Smaller gains (or losses): the squeeze bites where there's enough competition to contest the movable-collateralized top of the MFI book but not enough to serve the opaque bottom
- Quantifying each force and providing policy advice to foster gains is a central object of this project

Empirical Approach

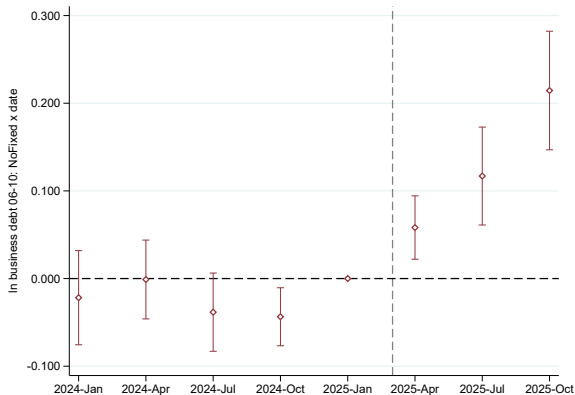
Exploit multiple sources of variation

- **Time**: the SIGM opens nationally in March 2025
- **Enforcement**: staggered connection of Superior Courts to the SIGM across judicial districts
- **Local competition**: pre-reform rival branch density in the borrower's city
- **Borrower characteristics**: size, number of lenders, formality, pre-reform collateral, productivity proxies
- **Lender type**: hard-information banks vs. relationship MFIs

Preliminary evidence

$$\ln \text{Debt}_{it} = \sum_{\tau \neq \text{Jan-25}} \beta_{\tau} \text{NoFix}_i \times \mathbf{1}\{t = \tau\} + \alpha_i + \delta_{s(i)t} + \eta_{k(i)t} + \theta_{c(i)t} + \varepsilon_{it}$$

- $\text{NoFix}_i = 1$ if borrower reports **no fixed collateral** in Jul-2024 to Jan-2025
- Business debt; natural-person borrowers
- Fixed Effects: borrower; size-bin, industry and district $\times$ quarter
- SE clustered by district; 95% CI
- Consistent with verifiable and enforceable baseline
- Next steps: channels, competition effects, small firm performance



Conclusions and next steps

Conclusions and next steps

- The SIGM made movable collateral **public and enforceable**
- Unique setting for studying small firm credit access when collateral is **verifiable and enforceable**, and understanding lender incentives to **share information**
- First pass: after March 2025, small borrowers without fixed collateral expand **business credit** relative to collateralized borrowers
- **Next steps:** the credit registry shows **who borrowed**, not **who registered**
 - SIGM electronic registries: filing date, creditor, borrower identifier (RUC/DNI), asset type and description, etc.
 - Key to address how small business credit evolves after registry, which lenders register which borrowers, and whether information boosts competition or not